

Assignment 6 (Deadline: March 17)
Solutions will be uploaded on March 17

1. [Stata exercise] Use the first series dataset from the “Data Set” folder

- (a) Is this time series variable stationary? Explain.
- (b) Use Stata to draw ACF. What is your conclusion?
- (c) Estimate the y series using the AR(1) model
- (d) Check if the errors in the model (c) are white noise. [Hint: you can use residuals to represent the errors.]

2. Consider a linear regression model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$, $i = 1, 2, \dots, n$ with y_i being a dummy variable.

- (a) Show that if $E(u_i | x_{i1}, x_{i2}) = 0$, then $E(y_i | x_{i1}, x_{i2})$ is equal to $\Pr(y_i = 1 | x_{i1}, x_{i2}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$.
- (b) Show that the homoskedasticity assumption does not hold in this linear probability model.
- (c) Explain the estimated coefficients $\hat{y}_i = -0.1 + 0.08 \text{monthly_income} + 0.01 \text{educ}$, where y_i is 1 if one has bought a car, 0 otherwise, *monthly_income* (in \$1000 unit) and *educ* is years of education.
- (d) Calculate $\hat{y}_i$ if *monthly_income* is \$8,000 and *educ* is 16. Does this person have a car? Explain.
- (e) Calculate $\hat{y}_i$ if *monthly_income* is \$15,000 and *educ* is 16. Explain your result. Does this result make sense? If not, how would you solve the problem?

3. Consider a probit model $\Pr(y = 1 | \mathbf{x}) = \Phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5)$.

- (a) Derive the partial effect of x_1 on $\Pr(y = 1 | \mathbf{x})$, $\frac{\partial \Pr(y = 1 | \mathbf{x})}{\partial x_1}$.
- (b) Apply the probit model on the Morz.dta dataset. Report the MLE result and calculate $\Pr(y = 1 | \text{nwifeinc} = 20, \text{educ} = 10, \text{kidslt6} = 0, \text{age} = 30, \text{exper} = 10)$ by hand.
- (c) Calculate the partial effect of *exper* on $\Pr(y = 1 | \mathbf{x})$ for the group of women with *nwifeinc* = 20, *educ* = 10, *kidslt6* = 0, *age* = 30, *exper* = 10 by hand.